

Multiple Choice

1. **Answer:** B

Options: A) 6; B) 18; C) 36; D) 180

Note: Correct option: B - 18

2. **Answer:** B

Options: A) The standard deviation of the population is also zero.; B) The sample mean and sample median are equal.; C) The sample may have outliers.; D) The population has a symmetric distribution.

Note: Correct option: B - The sample mean and sample median are equal.

3. **Answer:** C

Options: A) 34; B) 2; C) 17; D) 3

Note: Correct option: C - 17

4. **Answer:** A

Options: A) 29; B) 28; C) 27; D) 26

Note: Correct option: A - 29

5. **Answer:** B

Options: A) -9; B) -14; C) 14; D) 9

Note: Correct option: B - -14

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '1'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: Actual answer is '-49'.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** We seek to use Vieta's formulas. To be able to apply the formulas, we multiply both sides by $(x-1)(x-5)(x-10)(x-25)$ to eliminate the fractions. This gives

$$(x-5)(x-10)(x-25)+(x-1)(x-10)(x-25)+(x-1)(x-5)(x-25)+(x-1)(x-5)(x-10) = 2(x-1)(x-5)(x-10)(x-25)$$

(Be careful! We may have introduced one of the roots $x = 1, 5, 10, 25$ into this equation when we multiplied by $(x-1)(x-5)(x-10)(x-25)$. However, note that none of $x = 1, 5, 10, 25$ satisfy our new equation, since plugging each one in gives the false equation $1 = 0$. Therefore, the roots of this new polynomial equation are the same as the roots of the original equation, and we may proceed.) The left-hand side has degree 3 while the right-hand side has degree 4, so when we move all the terms to the right-hand side, we will have a 4th degree polynomial equation. To find the sum of the roots, it suffices to know the coefficients of x^4 and x^3 . The coefficient of x^4 on the right-hand side is 2, while the coefficients of x^3 on the left-hand and right-hand sides are 4 and $2(-1-5-10-25) = -82$, respectively. Therefore, when we move all the terms to the right-hand side, the resulting equation will be of the form

$$0 = 2x^4 - 86x^3 + \dots$$

It follows that the sum of the roots is $86/2 = 43$.

Note: Students should show all steps in their mathematical solution.

12. **Answer:** 6.68%. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key